
You Only Look Again: Self-Supervised Foveation

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1. Introduction

1.1. Motivation

With the recent strides of deep learning in computer vision, the literature has been pushing towards architectures hyper-optimized for accuracy, with not enough attention given to the efficiency of said models. The search for efficiency has largely been relegated to theory, and non-agnostic architectural improvements. On the other hand, the brain still remains a computational mystery at large, with its high adaptability and minimal power requirements remaining unsurpassed. Although, during the infancy of neural networks, inspiration was often drawn from the brain, this does not seem to be the case with the current state of the literature. These modern, deep models have far departed from any semblance of how the brain itself operates, it cannot be denied that there are large improvements to be made in both efficiency and robustness by drawing inspiration from the mind and its operations.

The problem of image classification appears to be solved at large with bulky CNNs and Attention-Based models. It can be argued however that these models are computationally expensive, and it is the purpose of this project to explore one biologically inspired avenue of selective processing to reduce the massive overhead costs inherent with these modern deep neural networks. Image Classifiers typically work by processing the entirety of the image at once. This is a massive waste of computational resources since we do not need every pixel in the image to inform us of its given class label. Instead, drawing inspiration from how humans recognize objects, we see that we need only focus (or rather foveate) our attention around a few key pixels of the image to decide on its class label.

1.2. Description

Such a foveation specialized model has already been proposed by (Cho et al., 2023), and this project aims to bolster the performance of their model by employing self-supervision during training to aid the model to be able to recover the original image from the compressed foveated version. We hope that this will improve performance by helping the encoder recover or 'hallucinate' lost detail about the periphery.

2. Related Work

Their approach naturally integrates both top-down and bottom-up attention, and it is this very model that this project will aim to improve. Their model first applies a space-variant Foveal Cartesian Geometry (FCG) sampling to compress each 256×256 image into an 81×81 “foveated” patch (10 of pixels) centered at the image midpoint, which it feeds through a standard CNN (batch-norm + ReLU + global average pooling) that outputs both class logits and a bottom-up attention map (the mean over the last conv-layer activations). The peak of that saliency map determines a single saccade: the sampling grid is shifted to that most salient location, a second foveated crop is taken, and the same CNN processes this “zoomed-in” view to yield a refined prediction. Top-down attention via Class Activation Maps further enables goal-driven fixations, and this two-glimpse pipeline achieves near-full-resolution accuracy with roughly tenfold fewer FLOPs and several-fold lower memory usage.

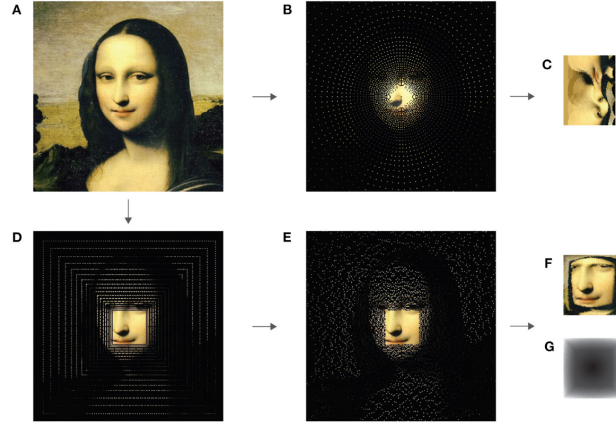


Figure 1. FCG foveation, used in our model (A→D→E→F). Log-polar transform (A→B→C) is provided for comparison. Notice that (B), (D) and (E) illustrate coordinates of the sampled pixels, represented by their true RGB value. The black background is meant to provide contrast for better visualization and is discarded during foveation. (A) Original image. (B) Log-Polar sampling. (C) Log-polar transform. (D) FCG sampling. (E) FCG sampling with jitter. (F) FCG foveated image. (G) A channel indicating sampled pixel positions (Lukanov et al., 2021)

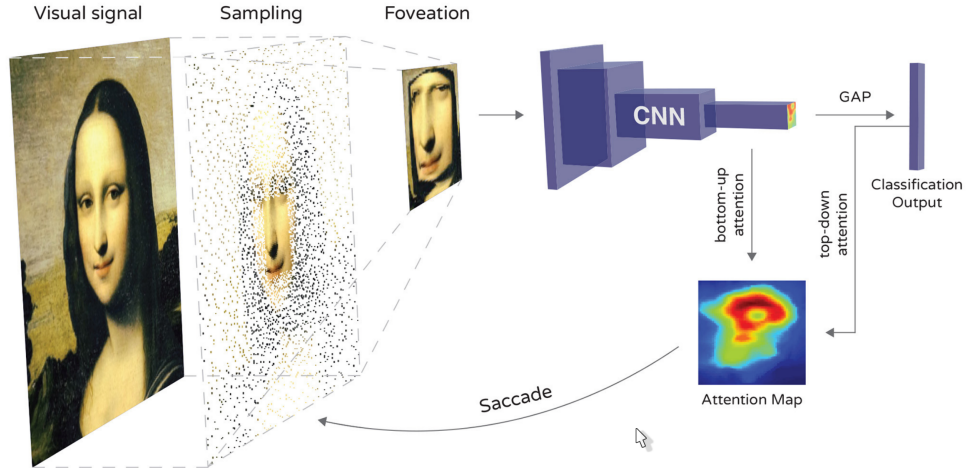


Figure 2. The baseline foveation model proposed by (Lukanov et al., 2021)

3. Key Idea (YOLA)

This project proposes to augment the two-glimpse foveated-vision model with a self-supervised training augmentation in which the shared CNN encoder learns to reconstruct the high-resolution original images from only the foveated patches. Concretely, during training we siphon the encoder’s features into a variety of decoders for the purposes of reconstruction. We will use Mean Squared Error to gauge the reconstruction quality. We use a total of two decoders and three losses to achieve our goal. The first and main decoder consists of layers of PixelShuffle to map the encoder outputs back to the Pixel Space. The PixelShuffle Decoder’s outputs are compared in the pixel space and the perceptual space using a frozen VGGNet13’s intermediate layers to enhance reconstruction. A second and more lightweight decoder serves to aid the flow of gradients back into the encoder, this decoder’s outputs are only evaluated in the pixel space. Our hope is that this will guide the encoder towards a better local minima in the loss space, by helping it recover lost detail from the foveated images.

We lovingly dub our model YOLA; You Only Look Again.

4. The Experiment

The experimental setup was simple; we implemented the Foveation model from scratch which served as the baseline for this experiment. Modifications were subsequently made to this baseline model to incorporate a self-supervised reconstruction loss, the details of which we discuss in the following sections.

4.1. Model Details

The models take in an input strictly of size [batch, 3, 256, 256]. These images are then passed through to a foveation layer, which down-samples these inputs to [batch, 3, 81, 81]. For the first glimpse (or rather look) the foveation layers foveate on the centers of the batch of images. These foveated images are then passed onto a CNN based backbone encoder which transforms them into [batch, 256, 3, 3]. These features are used for computing the logits for the first look and computing the saliency map to guide the second look. For YOLA, the reconstruction guided model, we pass these features onto two decoders: (i) a heavy PixelShuffle decoder that maps the features back into the pixel space using a pixel loss and perceptual loss in VGGNet’s mid-feature space, and (ii) a lightweight auxiliary decoder to aid the flow of gradients back to the encoder. A weighted combination of the saliency map (bottom-up) and the the Class Activation Maps (top-down) flowing from the logits determine the location of the next foveation. For the second glance, the encoder features are simply passed onto the logits, and no further computations are done.

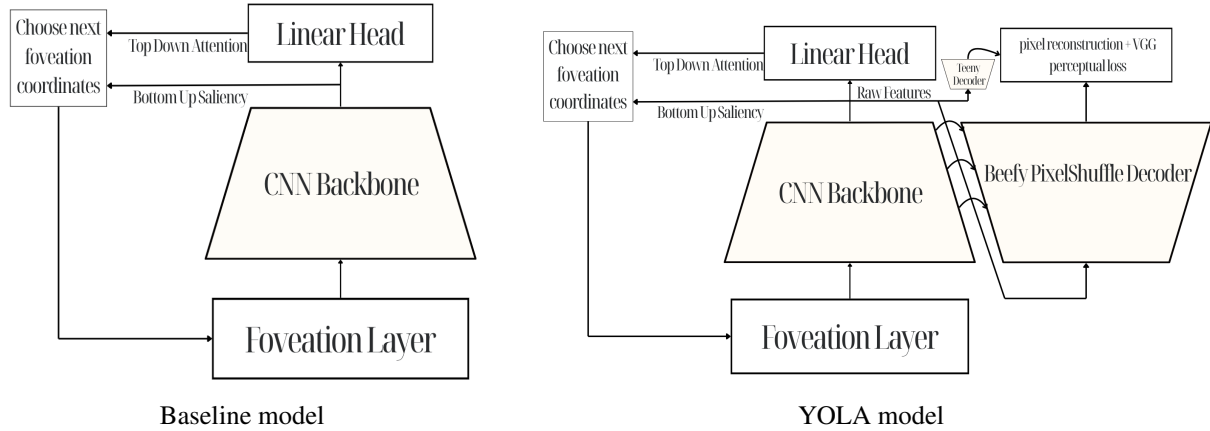


Figure 3. Architecture comparison between the Baseline Model and the Self-Supervised YOLA

4.2. Training Details

We trained the model for several training cycles on the Caltech-101 dataset, however we settled on the following training hyperparameters; we trained both the baseline foveation model and YOLA using an ADAM optimizer, with a learning rate of $1e - 4$ and a weight decay of $1e - 6$ for a total of 50 epochs. The perceptual and pixel reconstruction loss for YOLA were weighted by relatively high constants to emphasize their importance during learning which yielded the following loss term for YOLA:

$$L_{YOLA} = L_{cls} + L_{aux} + 3 * L_{perceptual} + 3 * L_{pixel}$$

During testing, we only record the classification losses for YOLA, and zero out reconstruction loss terms.

5. Results and Evaluation Criterion

5.1. Evaluation Criterion

We benchmarked both the models on a 20% test split of the Caltech-101 dataset, and evaluated their performances by simply considering their classification accuracies on the data. As a more subjective evaluation, we also paid attention to the quality of the reconstructions that YOLA produced through the decoder, which we will discuss in detail below.

5.2. Results

Across several training cycles on the Caltech-101 dataset, YOLA yielded better classification accuracies on the test dataset compared to the baseline model.

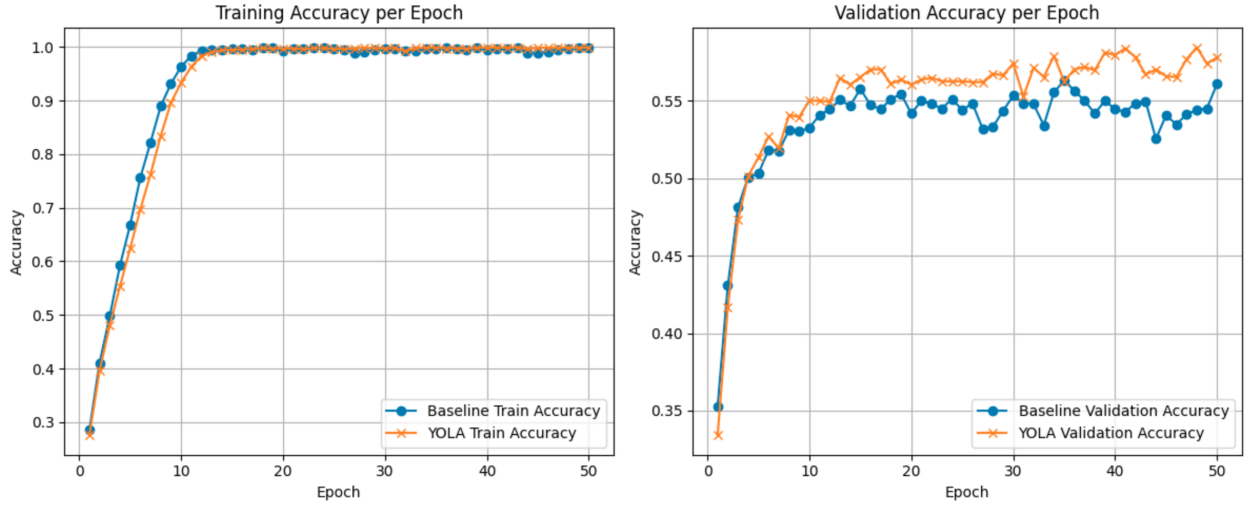


Figure 4. The training and validation accuracies of the baseline model and YOLA

As can be garnered from 4, we see that despite both the models effectively saturating on the training data, YOLA displays a strong immunity to overfitting, and displays more robustness on the validation dataset. The best reported validation accuracy for the baseline model is 56.31% while YOLA achieves 58.44%. It is also interesting to remark that the models begin to saturate/overfit on the 10th epoch, and on the validation dataset, both models display comparable performance up till the 10th epoch. Post 10th epoch however, YOLA takes over and remains performant compared to the baseline, which suggests that the self-supervised training routine made it less liable to overfit. It is important to remark here, that YOLA contains no extra components during the inference stage, and has the same architecture as the baseline. The reconstruction decoders only function during training, so effectively we have kept an exact parameter count while bolstering performance.

6. Discussion

Having settled the superior performance of our proposed model, we now turn our attention towards analyzing our proposed model. We first discuss the quality of reconstructions produced by the decoder at the inference stage to gather some insights about the model.

We remark and admit with figure 6 as our evidence that the reconstructions are not up to expectations. The decoder is unable to recover any color information or detailed features. We conjecture this is because classification and reconstruction are inherently competing tasks. The model was not able to recover much

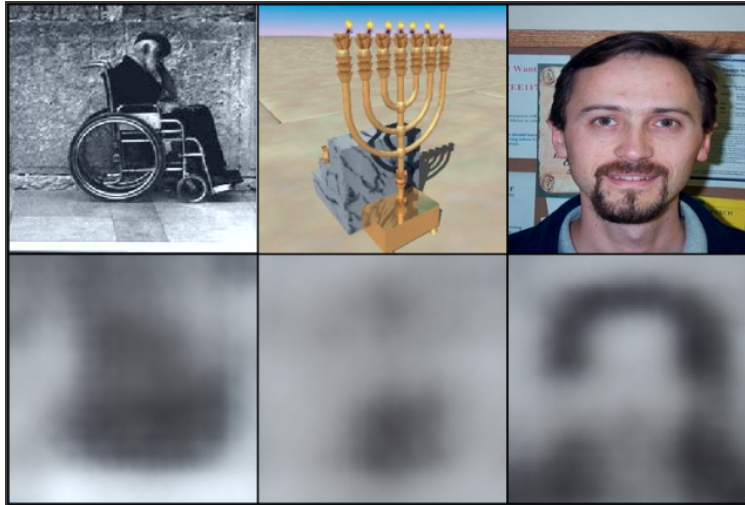


Figure 5. YOLA's decoder reconstructions for the Caltech-101 dataset

information about the periphery to bolster accuracies, however it seems that the act of adding a self supervision loss term guided the encoder to a more robust minima.



Figure 6. YOLA's decoder reconstructions for the MLRS dataset; a satellite imagery classification dataset where the model appears to recover color information to a degree

There were several methods that could have improved the reconstruction fidelity and colors, however these methods would have detracted from our purpose of forcing the encoder to learn better representations. For example, the addition of a GAN to discriminate the reconstruction outputs would have greatly improved the images. However, this would not have benefited the encoder itself, which happens to be our fundamental goal.

In essence our addition of self supervision appears to have worked, although not for the reasons we had hoped for. We also remark that our model did not achieve exceptionally superior performance on the MLRS dataset (beating the baseline by 0.1%), the reasons for which we did not have the time to investigate.

Although empirically this project appears a success, we are not satisfied with the quality of our results. For example, we could not explore in detail the increase in accuracy granted by the second look. The model

appears to have a substantial bump in second look accuracy during training, however for some innane reason, the first and second look accuracies appear to be equal, down to the 10th decimal during testing. This may be a bookkeeping error, however timely constraints did now allow us to resolve this issue properly despite our best efforts. Another point of contention that the reader may notice is that our encoder’s feature outputs have a spatial dimension of $(3, 3)$, which seems too small to predict good second look foveation coordinates when reversed back up to the image coordinates. We intended to modify our architecture to fix this and explore the generated saliency maps, however trainings took far too long given our current resources.

There remains much to be explored and there is plenty of room for improvement, including a learnable foveation layer using spatial transformers, combining top down and bottom up attention streams using more sophisticated and learnable methods, and simulating residual neuron activations for the second glimpse by means of a RNN-esque hidden connection that only unfolds once.

References

- Cho, S., Lee, J., and Kim, D. Hyperbolic vae via latent gaussian distributions, 2023. URL <https://arxiv.org/abs/2209.15217>.
- Lukanov, H., König, P., and Pipa, G. Biologically inspired deep learning model for efficient foveal-peripheral vision. *Frontiers in Computational Neuroscience*, 15, 2021. URL <https://api.semanticscholar.org/CorpusID:244464941>.